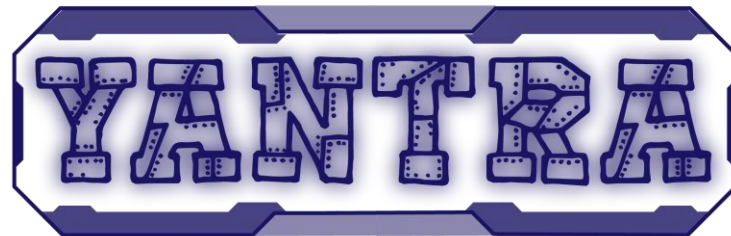
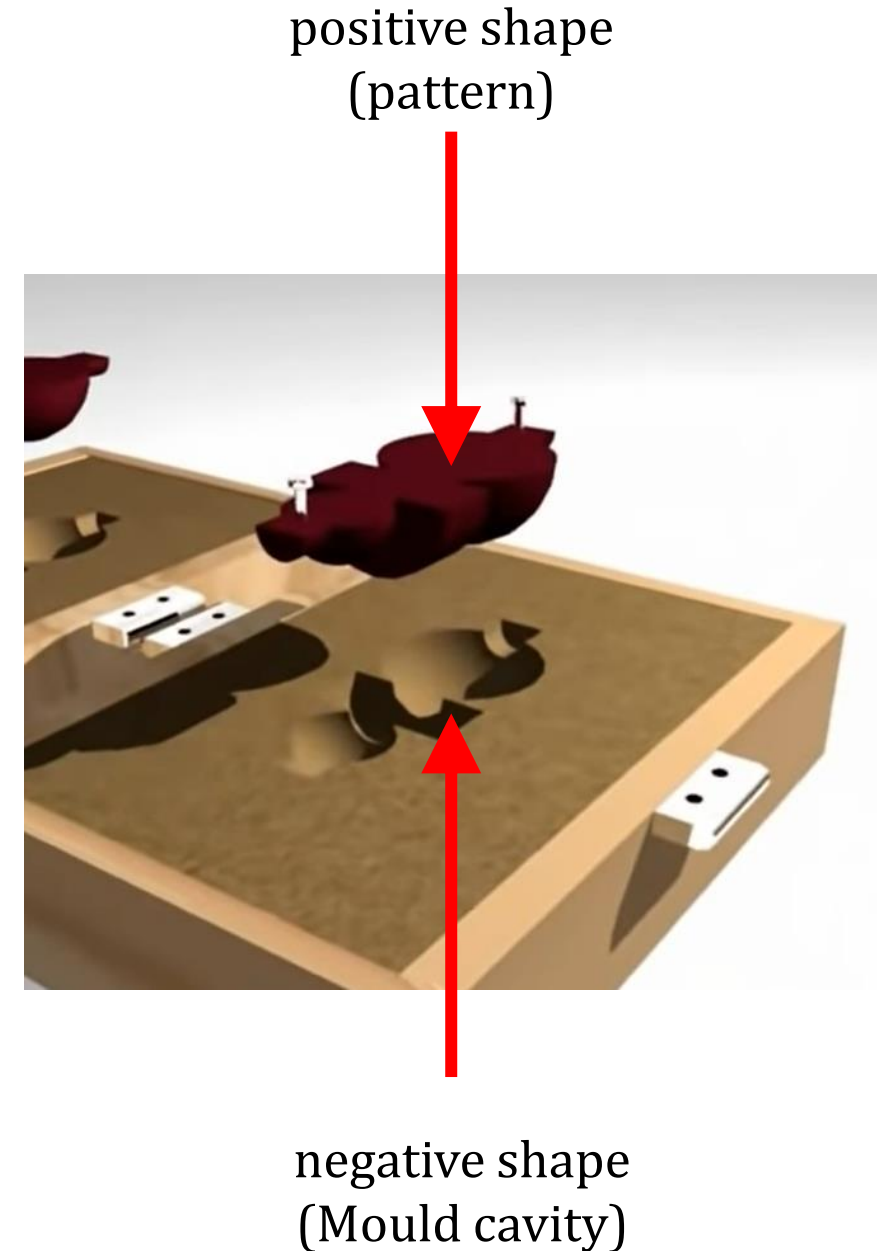


Rapid tooling

Introduction



- Rapid tooling refers to **mold cavities** that are either directly or indirectly fabricated using rapid prototyping (RP) techniques.
- **Soft tooling** can be used to inject multiple **wax or plastic** parts using conventional injection-molding techniques.



Conventional tooling

- **Traditional** hard-tooling patterns are fabricated by machining either tool **steel or aluminum** into the **negative shape** of the desired component.
- Steel tools are very expensive, yet typically **last indefinitely**, building **millions** of parts in a mass-production environment. Aluminum tools are less expensive than steel, although sometimes still \$10,000 or more, and are used for lower production quantities up to several hundred thousand parts.

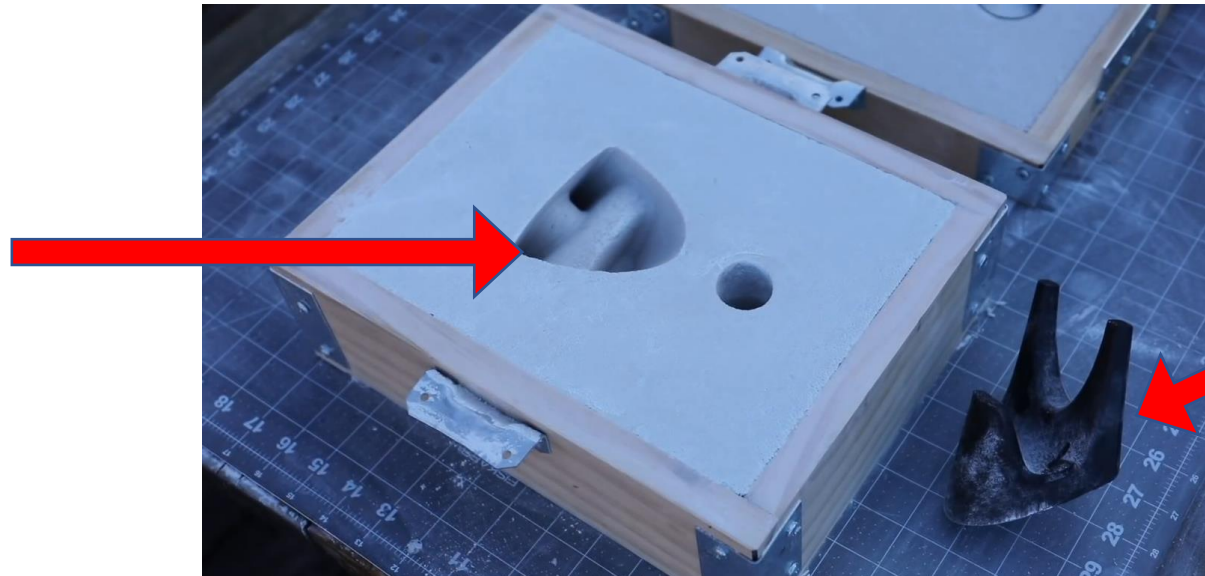


Traditional injection
molding tools

Rapid tooling	Conventional tooling
Tooling time is less i.e. 1/5th of conventional tooling.	Tooling time is high than rapid tooling.
Tool cost is less than conventional tooling.	Tool cost is high than Rapid tooling.
Tool life is less than conventional tooling.	Tool life is more than Rapid tooling.
Tolerance is wider than conventional tooling.	Tolerance are narrow than rapid tooling.

- Process can create metal tools directly from CAD files using RP Technique.
- Rapid tooling is divided into two types:
 - (i) Direct tooling, (ii) Indirect tooling- secondary process.
- In direct method, the fabricated part by RP machine itself is used as tool.
- In indirect method the part fabricated by RP machine is used as pattern in a secondary process.
- The resulting part from the secondary process is used as the tool.

Direct tooling
mold is made



Indirect tooling
Pattern is made



RAPID TOOLING

Classification of Rapid Tooling Processes

- **Indirect method**

1. Silicon rubber tooling
2. Spray metal tooling.
3. 3Dkeltool.
4. Composite tooling(Aluminium filled epoxy tooling)
5. Cast kirksite

Direct method

1. Direct AIM (ACES Injection Moulding)
2. Direct quick cast process
3. Direct method laser sintering
4. Copper polyamide,
5. Sand casting tooling

Direct Rapid Tooling

- Pros
 - Faster production
 - It involves fewer steps and resources to make
 - You can produce multiple prototypes
 - Flexible
- Cons
 - Less durable
 - Prone to minor errors
 - Not suitable for complex designs
 - Increase in cost

Indirect tooling

- Pros
 - It makes durable tools
 - More cost-effective
 - You can create hard or soft tools depending on your requirement
 - It produces tools with less discrepancies and variations
 - You can use it to experiment with different materials
- Cons
 - It is more time-consuming
 - Entails additional costs
 - It needs high quality materials to improve its durability
 - It does not fit simple designs and is more appropriate for complex designs

RAPID TOOLING Indirect Method

- **Soft Tooling**

- It can be used to intake multiple wax or **plastic parts** using conventional injection moulding techniques.
- It produces short term production patterns.
- Injected wax patterns can be used to produce castings. Soft tools can usually be fabricated for ten times less than a machine tool.



- **Hard Tooling**

- **Patterns** are fabricated by machining either tool **steel or aluminum** into the negative shape of the desired component.
- Steel tools are very expensive yet typically last indefinitely building millions of parts in a mass production environment.
- **Aluminum tools** are less expensive than steel and are used for lower production quantities.

